Supplementary Material: What Do Fairness Mitigations Do to Intersectional Gaps in Cardiovascular Risk Prediction?

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This supplement contains the extended methods, results and discussion of the main paper, additional robustness checks, an MLP, convergence checks, an exponentiated-gradient reference and a UCI replication. Table numbers (S1, S2, ...) and section letters match the references in the main paper. Section references of the form "Sec. III-A" inside copied text refer to Section A here.

A Extended methods and experimental setup

A.1 Federated fairness-penalized models

Client k holds D_k with n_k samples and minimizes

$$L_k = \frac{1}{n_k} \sum_i \ell(y_i, \hat{y}_i) + \frac{\mu}{2}\|w\|^2 + \lambda R_{\text{fair}}, \tag{1}$$

with binary cross-entropy ℓ, $\mu = 10^{-4}$, and a covariance-based DP proxy [2]

$$R_{\text{fair}} = \sum_{g=1}^{G} \big(\text{Cov}(s_g, z)\big)^2, \tag{2}$$

where s_g indicates group g and z is the logit. Squaring is necessary: signed covariances of one-hot indicators sum to zero. Age and sex are input features. Models: logistic regression (7 inputs) and a one-hidden-layer MLP (16 tanh units). FedAvg weights clients by n_k; E=3 full-batch local epochs.

EO penalty. For the EO comparison (Sec. B.2) the same squared covariance is computed separately within the positive and within the negative respondents of each client and the two terms are summed. It is a proxy for equal TPR and FPR; the schedule is stiffer (300 rounds at 0.15 for $\lambda \leq 8$, 600 at 0.075 for λ=16, 1200 at 0.0375 above).

Penalty groups. *Intersectional*: G=4 cells (age $\geq$ 55 vs. $<$ 55) $\times$ sex. *Sex-only*: G=2. *Age-only*: G=2. *Marginal-both*: penalties on the sex and the age indicators (G=4 overlapping columns), the natural way to "audit both attributes" without crossing them.

Pre-specified masking criterion. Before running the age-marginal comparison we fixed (file `PREREG_marginal.txt` in the code) that "marginals suffice" would be claimed only if the sex-only model's age-marginal DP gap is at least 0.05 and at least twice that of the sex+age marginal model; otherwise the claim would be dropped.

Training budget. The penalty makes the objective stiff, so one learning rate does not suit all settings. Logistic regression uses T=100 rounds at rate 0.3 for intersectional penalties with $\lambda \leq 8$ and T=300 at 0.15 for λ=16; for the sex-only, age-only and marginal-both penalties, $\lambda \leq 4$ uses T=100 at 0.3, λ=8 uses T=200 at 0.15 (marginal-both: 300 at 0.1), and λ=16 uses 400 at 0.075 (marginal-both: 600 at 0.05). A guard retries with half the rate and twice the rounds if pooled training AUROC drops below 0.55 or scores are non-finite; it fired in 194 of 9,835 logistic-regression fits (2.0%, all at $\lambda \geq 8$, by arm: label-balance 80 fits (6.3% of its fits), naive penalty 38 (1.7%), group re-weighting 40 (2.6%; 19 more in the global-count variant, 24%), oversampling 11 (0.5%), EO penalty 6 (2.9%)) and every retry converged. Appendix E compares a subset of settings (all arms in capped data; the headline arms at $\lambda \geq 8$ and the EO penalty in natural data) with half the rate and twice the rounds; the other runs are not checked. FedAvg is calibrated (ECE 0.006). MLP budget: Sec. H. λ is swept, not tuned.

A.2 Methods compared

Penalty-based variants use the intersectional DP penalty (G=4) at the same λ unless stated; oversampling and re-weighting are *combined with* that penalty, so their offsets measure the added effect.

1. **FedAvg** (λ=0); **Sex-only**, **Age-only**, **Marginal-both** and **Intersectional** ("naive") penalties as above.
2. **EO penalty** (Sec. A.1): label-conditional covariance penalty on the four groups, used only in the comparison with group thresholds (Table S4).
3. **Oversample**: each client duplicates every (group, label) cell up to its largest cell, with Gaussian noise (5% of the cell's standard deviation) on age. A simplified variant of FedIDA's oversampling [1].
4. **Controls**: label-only balancing by duplication; loss re-weighting by label, by group, or by (group, label) cell (weights $\propto 1/$cell count, mean 1); aggregation with original n_k; no noise. Group re-weighting is computed from *local* counts in each client by default; a *global* variant uses weights from the pooled group counts, which a server could compute by secure aggregation of four counts.
5. **Shrinkage**: local covariance blended with the pooled esti-

mate, $\alpha = n_{k,g}/(n_{k,g} + 20)$ (heuristic).

6. **Group thresholds** (post-processing on FedAvg scores): *equal selection* (each group's threshold gives the overall selection rate) and *equal TPR* (each group's threshold gives the overall TPR) [4], fitted on the training scores or on a held-out 20% validation split of each client ("val. split"; model trained on the other 80%). They use group membership at decision time.
7. **Centralized matched control**: all clients pooled into one, with the same number of gradient steps ($T \times E$), same penalty and variants.

An exponentiated-gradient reduction [6] is reported only as an illustrative reference (Appendix F).

A.3 Metrics and equal-AUROC comparison

$\mathrm{DP} = \max_g P(\hat{Y}{=}1|g) - \min_g P(\hat{Y}{=}1|g)$; EO is the larger of the max–min TPR gap and FPR gap, over the 2 sex groups or the 4 age×sex groups ("sex", "int"). DP is a parity *audit*, not a clinical target: base rates genuinely differ (2.2%–18.0%), and forcing equal selection rates sacrifices sensitivity for older men. Two decision rules. (q) *Prevalence-matched*: classify the top fraction of held-out scores equal to the training prevalence (9.3%). (fx) *Deployable*: a fixed score threshold from the training scores at the training prevalence. Both are monotone in the score, so they are unaffected by an intercept shift. Because a single operating point can mislead, Sec. B.3 repeats the audit at other selection rates and fixed risk cut-offs and reports group-wise AUROC and calibration. We also report sensitivity, PPV, the smallest group PPV, and calibration (10-bin ECE, before and after an intercept shift that matches the mean score to the nominal prevalence).

For each method we sweep λ over a grid (Table S6 for the fine grid 0.1–16; $\{1, \ldots, 16\}$ for the core suites). *Within each seed*, we trace the naive penalty's frontier (AUROC, gap) by linear interpolation and compute each method's offset from it at the same AUROC, averaged over in-range λ; negative = smaller gap at equal accuracy. Because the offset averages over the in-range λ values, it depends on the grid: we therefore also report offsets at each λ (Table S12) and under the fine grid (Sec. B.4). Intervals are percentile bootstraps over seeds (10,000 resamples). p-values are two-sided sign-flip permutation tests of a zero mean offset (exact for ≤ 16 seeds, so the smallest attainable p is $2/2^n$: 0.002 for 10 seeds, 0.0625 for 5; Monte Carlo with 200,000 sign flips for 20 seeds). The primary family is **12 logistic-regression tests**, Holm-corrected: {oversample, label-balance, group re-weight} × {DP, EO} × {natural, capped}, each on one dense λ grid with 20 seeds per scenario (the natural suites were extended from 10 to 20 so that the Holm floor is not binding). With 10 seeds the smallest possible Holm-adjusted value would be about $12 \times 0.00195 \approx 0.023$, so "not significant" at that size does not mean "absent". MLP tests (5 natural, 10 capped seeds) and everything else are exploratory and reported with unadjusted intervals.

Table S1: Number of seeds per experiment suite (seeds 42–61 at most; natural = full state samples, capped = 150 respondents per client).

Suite	Natural	Capped
Core, centralized control, controls, dense grid (λ=0.5, 2)	20	20
Thresholds, operating points, group-wise metrics	10	20
EO penalty vs. thresholds	10	20
Leave-states-out, age cut-offs	10	20
Fine λ grid	5	20
Age-marginal gap (Table S3)	5	20
Dirichlet partitions (federated and centralized)	10	–
MLP	5	10
Convergence checks (production vs. half rate)	5 (select arms)	5

A.4 Experimental setup

BRFSS 2023 [7]. After complete-case restriction, 317,091 respondents from 52 states/territories; each state is one client (median 4.6K respondents). Features: age (imputed, top-coded at 80), BMI, high blood pressure, high cholesterol, diabetes, smoking, sex. Outcome: ever-told coronary heart disease or myocardial infarction (9.3%); age×sex prevalence 2.2% (F, < 55), 2.8% (M, < 55), 9.9% (F, ≥ 55), 18.0% (M, ≥ 55). Primary protocol: 85/15 random split per seed. *Natural*: full state samples. *Capped*: each client subsampled to 150 respondents to emulate hospital-scale clients. Seeds are 42–61 at most; the number per suite is in Table S1.

A state stands in for a care site, not a hospital. *Leave-states-out*: 15% of states are held out entirely (10 natural / 20 capped partitions). *Heterogeneity.* Real states are close to IID: the SD of client prevalence is 0.019 (range 0.049–0.150) and the mean total-variation distance between a client's age×sex mix and the pooled mix is 0.059. To probe stronger heterogeneity we repartition the same training data into 52 clients by sampling each (group, label) cell across clients from a Dirichlet distribution: α=1 (prevalence SD 0.071, mean TV 0.282) and α=0.3 (SD 0.144, TV 0.420). UCI Heart Disease [8] (851 patients, four sites) is a small secondary check (Appendix G).

Training budgets. Under-trained earlier runs (25, then 80 rounds) changed penalized-model results and the MLP offsets; all numbers here use the stiffness-aware schedule above, and the earlier "monotone EO" and "MLP is immune" conclusions are withdrawn.

B Extended results

B.1 Auditing one attribute misses the intersection; auditing both marginals does not

On natural BRFSS at λ=8 (Tables S2, S3; this subsection: 5 natural and 20 capped seeds, except where stated) the sex-only model cuts the sex DP gap by 87% (0.111→0.015) and the sex EO gap by 93% (0.272→0.019), but the age×sex gaps by only 33% (DP 0.255→0.171) and 28% (EO 0.462→0.332), at AUROC 0.798→0.789. A sex-only audit would pass this model on DP (0.015) although its intersectional DP gap is 12× larger.

Table S2: Main BRFSS results at λ=8 (logistic regression; mean ± std. over 20 seeds per scenario; rule (q)). Lower is fairer. "int" = 4 age×sex groups.

Method	AUROC	DP_{sex}	DP_{int}	EO_{sex}	EO_{int}
Natural (52 states, ∼6K each; 20 seeds)					
FedAvg	0.798±0.003	0.111±0.002	0.255±0.002	0.272±0.014	0.462±0.010
Sex-only	0.789±0.003	0.014±0.002	0.171±0.002	0.019±0.009	0.332±0.007
Naive inter.	0.680±0.004	0.002±0.001	0.033±0.004	0.037±0.012	0.222±0.017
Oversample	0.706±0.004	0.003±0.002	0.046±0.004	0.044±0.011	0.210±0.019
Shrinkage	0.680±0.004	0.002±0.001	0.033±0.004	0.037±0.012	0.222±0.017
Capped (n=150 per client; 20 seeds)					
FedAvg	0.797±0.003	0.108±0.015	0.251±0.014	0.261±0.038	0.453±0.019
Sex-only	0.787±0.004	0.012±0.005	0.167±0.004	0.014±0.011	0.325±0.008
Naive inter.	0.681±0.005	0.004±0.002	0.048±0.014	0.034±0.017	0.171±0.065
Oversample	0.700±0.008	0.010±0.007	0.053±0.012	0.035±0.023	0.190±0.048
Shrinkage	0.681±0.005	0.004±0.002	0.046±0.012	0.033±0.016	0.178±0.063

Table S3: Masking at λ=8 (fine-grid suite; logistic regression, rule (q)). Reductions are relative to FedAvg. "Marginal" penalizes the sex and age indicators together; "Intersectional" the four age×sex cells. DP_{age}: old vs. young selection-rate gap.

Penalty	AUROC	DP_{sex}	DP_{age}	DP_{int}	ΔDP_{int}	EO_{int}	ΔEO_{int}
Natural (5 seeds)							
FedAvg	0.798	0.112	0.152	0.254	–	0.459	–
Sex-only	0.789	0.015	0.154	0.171	33%	0.333	28%
Age-only	0.695	0.131	0.044	0.175	31%	0.352	23%
Sex+age marginal	0.677	0.002	0.024	0.031	88%	0.232	49%
Intersectional	0.681	0.002	0.026	0.032	87%	0.228	50%
Capped (20 seeds)							
FedAvg	0.797	0.108	0.150	0.251	–	0.453	–
Sex-only	0.787	0.012	0.151	0.167	33%	0.325	28%
Age-only	0.694	0.118	0.036	0.155	38%	0.336	26%
Sex+age marginal	0.676	0.004	0.034	0.043	83%	0.182	60%
Intersectional	0.681	0.004	0.039	0.048	81%	0.171	62%

Capped data agree (DP_{int} 0.251→0.167, EO_{int} 0.453→0.325). An age-only penalty leaves the sex gap in place (DP_{sex} 0.131). The masking is visible on the age marginal itself: the sex-only model's age DP gap is 0.154 against 0.152 for FedAvg (natural, 5 seeds; 0.151 against 0.150 capped, 20 seeds), whereas the marginal-both model reaches 0.024 (0.034 capped); the prespecified threshold (Sec. A.4) is met in both scenarios.

Penalizing the sex and age marginals *together* closes the intersectional gap almost as well as penalizing the four cells: DP_{int} 0.031 vs. 0.032 and EO_{int} 0.232 vs. 0.228 (natural), 0.043 vs. 0.048 and 0.182 vs. 0.171 (capped). At equal AUROC against the intersectional frontier, the marginal-both offsets are DP -0.002 $[-0.003, -0.001]$ and EO $+0.007$ $[+0.003, +0.012]$ (capped, 20 seeds) and DP $+0.000$, EO -0.005 (natural, 5 seeds; the permutation p cannot go below 0.0625 with 5 seeds). With two binary attributes, then, the masking is a failure of auditing *one* attribute, not of marginal fairness: an audit of both marginals would have flagged the sex-only model. Whether crossing matters with more or finer attributes is untested here and is the setting where [5] predicts it should.

B.2 What penalties cost, how strength matters, and group thresholds

Cost. Intersectional penalties close most of the DP gap, but AUROC falls from 0.798 to 0.680 (natural, λ=8; 0.681 capped), and sensitivity at the deployable rule (fx) from 0.310 to 0.210 (natural; 0.310 to 0.215 capped; Table S5). Group-wise AUROC falls in every group, most for the older groups (older women/men 0.72/0.70 to 0.67/0.65; younger 0.80/0.79 to 0.78/0.76; Table S9). The cost is not due to federation: FedAvg and its centralized control have the same AUROC (0.797).

Strength. The EO gap is U-shaped in λ (Table S6). For the intersectional penalty (natural) EO_{int} is 0.459 at λ=0, 0.371 at 0.1, 0.100 at 0.5, 0.105 at 1, then 0.218 at 4 and 0.228 at 8; capped: 0.453, 0.352, 0.129 (0.5), 0.126 (1), 0.165 (4), 0.171 (8). The rise from λ=1 to 8 is $+0.123$ $[+0.117, +0.128]$ natural (5 seeds) and $+0.045$ $[+0.025, +0.064]$ capped (20 seeds). DP_{int} decreases monotonically. At λ=8 the penalty selects younger women at TPR 0.40 against an overall sensitivity of 0.21 at the 9.3% operating point, which is consistent with equal selection rates over-serving the low-prevalence groups, but we do not test that explanation. A mechanism consistent with the U-shape, from the intersectional DP penalty on 5 natural seeds (correlational, not a causal test): the age coefficient of the model is $+0.766$ at λ=0, -0.001 at 0.5 (the EO minimum), -0.111 at 1 and -0.24 at 8 (sex coefficient $+0.272 \rightarrow +0.024 \rightarrow -0.055$), so the penalty first removes the model's dependence on age and then over-corrects past it, which raises the EO gap. At the EO-optimal λ=0.5 the cost is smaller than at λ=8: AUROC 0.751 against 0.797 for FedAvg. "Monotone" statements about EO hold only for $\lambda \geq 1$, and comparing mitigations at a single large λ therefore overstates the penalty's EO gap.

Group thresholds. Thresholds fitted on FedAvg scores leave the ranking, hence AUROC (0.797), unchanged by construction; they change the operating point. Under the deployable rule (Table S7), *equal TPR* gives EO_{int} 0.087 (natural) and 0.142 (capped). Against the penalty at the

same rule: natural, 0.102 at λ=1 (paired difference -0.015 $[-0.031, +0.000]$), 0.220 at λ=4 (-0.133 $[-0.144, -0.122]$) and 0.232 at λ=8 (-0.145 $[-0.155, -0.135]$); capped, 0.125 at λ=1 ($+0.018$ $[-0.016, +0.049]$) and 0.168 at λ=8 (-0.026 $[-0.061, +0.009]$). So thresholds beat a strong penalty on EO in natural data only, and tie with a weak penalty everywhere; but the weak penalty (AUROC 0.716–0.719) already loses sensitivity (0.231 vs. 0.310) and PPV (0.233 vs. 0.270). When the equal-TPR thresholds are fitted on a validation split of about 30 respondents per client (capped), EO is worse (0.237; $+0.095$ over the training-fit thresholds) because few positives determine four thresholds; in natural data the variant matches (0.090 vs. 0.087). *Equal selection* gives DP_{int} 0.005 / 0.015 (natural / capped) against 0.032 / 0.046 for the penalty at λ=8, but lowers sensitivity (0.310→0.236) and the smallest group's PPV (0.238→0.114), and leaves EO at 0.268. Unconstrained FedAvg at the prevalence-matched threshold never flags one group (smallest group TPR 0.000); equal selection raises it to 0.197 and equal TPR to 0.283 (natural). Each rule optimizes one metric.

Against an EO penalty. The comparison above uses the DP-type penalty, which does not target EO. We therefore also trained a label-conditional covariance EO penalty (Sec. A.1; Table S4, same fixed-threshold rule, 10 natural / 20 capped seeds). Natural: equal-TPR thresholds have lower EO_{int} than the EO penalty at every strength tested, by -0.009 $[-0.012, -0.006]$ against λ=0.5 (0.087 vs. 0.096), -0.015 $[-0.037, +0.006]$ against λ=1, and -0.107 $[-0.123, -0.090]$ against λ=4, while keeping 0.067–0.112 more AUROC. Capped: the EO penalty is not distinguishable from thresholds on EO (paired differences -0.030 to $+0.022$, all intervals contain zero), but still costs 0.032–0.084 AUROC. As with the DP-type penalty, EO rises again at strong penalties ($\lambda \geq 2$ natural), and individual seeds at $\lambda \geq 8$ differ in EO by up to 0.07 between schedules, and two fits collapsed without the guard (Appendix E); the per-seed EO range at each λ is in Table S4, so we read high-λ EO-penalty numbers as unreliable rather than as a stable property of the method. Per-seed standard deviations of EO_{int} for the EO penalty are 0.003 (λ=0.5) and 0.022–0.032 (other λ) in natural data and 0.049–0.070 in capped data, so single-seed EO-penalty numbers are noisy at every strength except the weakest natural one. The defensible statement is that a post-hoc threshold matches or beats the EO penalty on EO at no AUROC cost, not that thresholds are better in general; it holds for one dataset and a covariance penalty, and not for a stronger in-processing method.

B.3 Operating-point sensitivity

Every gap above is measured at one operating point, the top 9.3% of scores. Table S8 repeats the audit at other selection rates. FedAvg's EO_{int} grows from 0.307 (5% selected) to 0.698 (20%) in natural data (0.300 to 0.664 capped), and the youngest group of women has TPR 0.000 at 5%, 9.3% and 15% (0.015 at 20%). With fixed risk cut-offs of 0.10, 0.15 and 0.20 FedAvg selects 35%, 22% and 14% of respondents (natural) with youngest-women TPR 0.125, 0.042 and 0.000. The penalized models' EO is not monotone in the selection rate (0.132, 0.232, 0.171, 0.151 at 5, 9.3, 15, 20% for the natural λ=8 penalty), and the comparison of oversampling with the penalty at equal λ also changes with the rate. In natural data oversampling has lower EO than the penalty at all four rates, but by 0.016 at 9.3% (0.216 vs. 0.232), 0.009 at 20% and 0.048 at 15% (0.142 vs. 0.151; 0.123 vs. 0.171); in capped data the ordering itself changes (oversampling is higher at 5, 9.3 and 20%, lower at 15%: 0.140 vs. 0.161). The models are not at equal AUROC, so this illustrates sensitivity to the operating point, not an effect. Our offset analyses use 9.3% and the deployable threshold, and the sign and size of EO effects should be read as specific to those points. Table S9 shows what the penalty does to group-wise discrimination and calibration: it lowers AUROC in all groups and shifts calibration-in-the-large from about zero to $+0.065$ (younger women), $+0.061$ (younger men), -0.004 (older women) and -0.084 (older men), i.e. it under-predicts risk for older men by 8 points while overpredicting for younger groups. Oversampling's calibration shift is a uniform prior shift of about $+0.3$ to $+0.4$ in every group.

Table S4: EO penalty versus group thresholds under the fixed-threshold rule (fx). Mean over 10 natural / 20 capped seeds; min TPR is the smallest age×sex group.

Method	AUROC	DP_{int}	EO_{int}	EO range (min–max)	min TPR
Natural (10 seeds)					
FedAvg	0.797	0.254	0.460	0.45–0.47	0.000
FedAvg + equal-TPR thr.	0.797	0.114	0.087	0.08–0.09	0.283
FedAvg + equal-sel. thr.	0.797	0.005	0.268	0.24–0.28	0.197
EO penalty λ=0.5	0.730	0.116	0.096	0.09–0.10	0.213
EO penalty λ=1	0.717	0.074	0.102	0.06–0.15	0.215
EO penalty λ=4	0.698	0.049	0.193	0.17–0.23	0.194
EO penalty λ=16	0.685	0.055	0.159	0.11–0.20	0.199
Capped (20 seeds)					
FedAvg	0.797	0.252	0.454	0.42–0.49	0.000
FedAvg + equal-TPR thr.	0.797	0.116	0.142	0.08–0.24	0.253
FedAvg + equal-sel. thr.	0.797	0.015	0.257	0.21–0.31	0.193
EO penalty λ=0.5	0.765	0.155	0.173	0.09–0.32	0.145
EO penalty λ=1	0.743	0.119	0.121	0.07–0.23	0.190
EO penalty λ=4	0.719	0.075	0.133	0.07–0.26	0.207
EO penalty λ=16	0.712	0.078	0.137	0.05–0.23	0.195

B.4 Oversampling: what it does, and what drives it

Primary estimates use one grid per scenario so the Holm family is not mixed across grids: the dense grid $\lambda \in \{0.5, 1, 2, 4, 8, 16\}$ in both scenarios, 20 seeds, naive frontier built from all six points (Fig. 2; Table S14, bottom rows; Table S11 gives the core-grid values). For logistic regression, oversampling lowers DP (-0.012 $[-0.014, -0.010]$ natural) and raises EO ($+0.059$ $[+0.054, +0.064]$) (per-seed range $+0.044$ to $+0.081$); both survive Holm correction over the 12 logistic-regression tests (adjusted $p < 0.001$). In capped data neither does: DP -0.006 $[-0.014, +0.002]$ (adjusted $p = 0.47$), EO $+0.020$ $[-0.004, +0.041]$ (adjusted $p = 0.45$). The core-grid natural EO value ($+0.072$) is larger because the naive EO curve is U-shaped and the chord from λ=1 to 4 lies below the naive points at λ=2, so chord-based offsets overstate (Table S12: natural dense per-λ offsets fall from $+0.085$ at λ=1 to $+0.038$ at

Table S5: Deployable rule (fx), λ=8, mean over seeds (10 natural, 20 capped) [95% CI over seeds]. Sens./PPV: overall; min PPV: smallest age×sex group. ECE: raw / after an intercept shift to the nominal prevalence (thresholds do not change scores).

Method	AUROC	DP_{int}	EO_{int}	Sens.	PPV	min PPV	ECE
Natural (10 seeds)							
FedAvg	0.797	0.254 [0.252, 0.255]	0.460 [0.456, 0.464]	0.310	0.312	0.238	0.006 / 0.006
+ group thr. (equal selection)	0.797	0.005 [0.004, 0.007]	0.268 [0.260, 0.275]	0.236	0.238	0.114	– / –
+ group thr. (equal TPR)	0.797	0.114 [0.112, 0.115]	0.087 [0.085, 0.089]	0.310	0.270	0.152	– / –
+ group thr. (equal TPR, val. split)	0.797	0.114 [0.109, 0.119]	0.090 [0.085, 0.097]	0.311	0.269	0.152	– / –
Sex-only penalty	0.788	0.171 [0.169, 0.172]	0.332 [0.330, 0.334]	0.290	0.293	0.184	0.007 / 0.007
Naive inter. penalty	0.679	0.032 [0.030, 0.033]	0.232 [0.223, 0.243]	0.210	0.213	0.122	0.007 / 0.007
Oversample	0.705	0.046 [0.044, 0.047]	0.215 [0.203, 0.228]	0.213	0.217	0.126	0.351 / 0.007
Group re-weight	0.679	0.033 [0.031, 0.034]	0.226 [0.214, 0.238]	0.212	0.215	0.124	0.011 / 0.006
Capped (20 seeds)							
FedAvg	0.797	0.252 [0.245, 0.259]	0.454 [0.445, 0.464]	0.310	0.309	0.217	0.006 / 0.006
+ group thr. (equal selection)	0.797	0.015 [0.012, 0.018]	0.257 [0.245, 0.270]	0.233	0.235	0.109	– / –
+ group thr. (equal TPR)	0.797	0.116 [0.110, 0.123]	0.142 [0.121, 0.164]	0.310	0.262	0.136	– / –
+ group thr. (equal TPR, val. split)	0.796	0.116 [0.099, 0.133]	0.237 [0.202, 0.273]	0.317	0.249	0.116	– / –
Sex-only penalty	0.787	0.167 [0.164, 0.171]	0.326 [0.321, 0.331]	0.290	0.290	0.138	0.015 / 0.016
Naive inter. penalty	0.681	0.046 [0.040, 0.052]	0.168 [0.141, 0.196]	0.215	0.221	0.118	0.017 / 0.017
Oversample	0.700	0.050 [0.044, 0.055]	0.190 [0.170, 0.210]	0.212	0.218	0.121	0.311 / 0.025
Group re-weight	0.681	0.049 [0.042, 0.055]	0.157 [0.128, 0.187]	0.217	0.221	0.117	0.016 / 0.017

λ=16). In capped data the per-λ offsets are mostly indistinguishable from zero; only λ=8 excludes zero ($+0.003$ to $+0.070$). At capped λ=0.5 the equal-AUROC offset is not estimable: the oversampled models (AUROC 0.759) lie above the top of the naive frontier (0.751) in 17 of 20 seeds. The equal-λ comparison (Table S10) does show a significant EO difference there ($+0.033$ $[+0.015, +0.052]$), the only capped point that does; it is diluted in the average because it is an equal-λ, not an equal-AUROC, estimate and the other capped points are null. The fine grid adding $\lambda \in \{0.1, 0.25, 0.5\}$ (Table S6) gives $+0.003$ $[-0.016, +0.020]$ capped and $+0.049$ $[+0.042, +0.055]$ natural, so the size of the average offset depends on the grid, and in capped data its sign does too.

The comparator matters. Table S10 compares at equal λ: the oversampled models have *lower* natural EO than the naive penalty for $\lambda \geq 4$ because they keep 0.02–0.03 more AUROC, and higher EO at $\lambda \leq 1$, where the naive penalty is near its EO optimum (naive λ=0.5: AUROC 0.751, EO 0.100). So "raises EO" holds at equal AUROC, not at equal penalty strength. A state-level cluster bootstrap that resamples test states with models fixed gives EO $+0.069$ $[+0.056, +0.082]$ (natural) and $+0.032$ $[+0.022, +0.042]$ (capped); it reflects geographic sampling of the test population only, not training variability.

Class rebalancing explains a large, setting-dependent share. Table S14 (exploratory). Aggregating with original n_k or removing the noise leaves the effect unchanged. Duplicating only to balance *labels* reproduces a share of the EO shift that depends on the setting: 49% on the dense natural grid ($+0.029$ against $+0.059$), 75% on the natural core grid ($+0.042$ against $+0.056$), and 90% on the capped dense grid ($+0.018$ against $+0.020$, both within noise of zero). The paired difference (oversample − label-balance) is $+0.030$ $[+0.027, +0.034]$ in natural data (DP -0.009) and $+0.001$ $[-0.011, +0.015]$ in capped data (DP $+0.002$), so in natural data a clear part of the shift is *not* explained by label balance. Under Dirichlet α=1 partitions label-balance gives about 90% (Table S16); at α=0.3 it exceeds oversampling (EO $+0.042$ against $+0.012$). Label-only loss weights (-0.006, $+0.030$ core grid) and (group, label) weights (-0.003, $+0.027$) behave similarly. Re-weighting by *group alone* is different: EO -0.011 $[-0.013, -0.009]$ (natural dense; adjusted $p < 0.001$) and -0.011 (capped dense; adjusted $p = 0.003$) with DP $+0.001$ to $+0.002$. These are gains of about 0.01 EO.

Its miscalibration is a prior shift. Oversampling raises ECE from 0.006 to 0.351 (natural) and 0.311 (capped) because the training prevalence moves toward 50%. One intercept shift to the nominal prevalence brings ECE back to 0.007 / 0.025, and it cannot change DP or EO, which depend only on score ranks under both decision rules. Calibration therefore does not explain the DP/EO shifts; it is a separate, fixable property that matters only if probabilities are used directly.

B.5 Is it specific to federation?

A centralized control with the same number of gradient steps (Table S13) gives oversampling DP -0.016 / -0.008 and EO $+0.052$ / $+0.025$ (natural, dense frontier / capped). Federation adds a small but significant amount in natural data (EO $+0.008$ $[+0.005, +0.010]$, DP $+0.004$ $[+0.004, +0.005]$) and nothing measurable in capped data (EO $+0.009$ $[-0.014, +0.031]$). On near-IID clients the oversampling effect is therefore a property of the mitigation, not of federation. Under skewed clients that changes (Sec. B.6): the oversampling EO effect shrinks federated and group re-weighting with local counts turns harmful, an effect that does not occur centrally and that a global-count variant removes. Federated training thus describes how the models are

Table S6: Fine penalty grid, $\lambda \in \{0.1, \ldots, 16\}$ (logistic regression, rule (q); mean over 20 capped / 5 natural seeds). EO of the intersectional and marginal-both penalties is U-shaped; sex-only stays high for all λ.

	Intersectional			Marginal-both			Sex-only		
λ	AUROC	DP_{int}	EO_{int}	AUROC	DP_{int}	EO_{int}	AUROC	DP_{int}	EO_{int}
Capped (20 seeds)									
0.1	0.789	0.222	0.352	0.778	0.202	0.273	0.794	0.201	0.378
0.25	0.774	0.191	0.249	0.751	0.139	0.131	0.792	0.186	0.354
0.5	0.751	0.138	0.129	0.719	0.083	0.127	0.790	0.178	0.341
1	0.719	0.084	0.126	0.703	0.059	0.154	0.789	0.174	0.333
2	0.704	0.062	0.148	0.689	0.048	0.177	0.788	0.171	0.329
4	0.691	0.053	0.165	0.680	0.042	0.194	0.788	0.169	0.329
8	0.681	0.048	0.171	0.676	0.043	0.182	0.787	0.167	0.325
16	0.676	0.049	0.160	0.673	0.044	0.172	0.786	0.164	0.312
Natural (5 seeds)									
0.1	0.791	0.228	0.371	0.781	0.210	0.294	0.795	0.200	0.375
0.25	0.777	0.199	0.272	0.753	0.122	0.100	0.793	0.186	0.354
0.5	0.752	0.122	0.100	0.719	0.079	0.094	0.791	0.179	0.344
1	0.719	0.076	0.105	0.703	0.046	0.193	0.790	0.175	0.337
2	0.703	0.046	0.195	0.689	0.038	0.217	0.790	0.172	0.335
4	0.689	0.037	0.218	0.681	0.033	0.227	0.789	0.171	0.333
8	0.681	0.032	0.228	0.677	0.031	0.232	0.789	0.171	0.333
16	0.677	0.030	0.235	0.674	0.030	0.234	0.789	0.171	0.333

Table S7: Deployable rule: equal-TPR thresholds on FedAvg versus the intersectional penalty at several λ. Mean over 10 natural / 20 capped seeds.

Method	AUROC	DP_{int}	EO_{int}	Sens.	PPV
Natural (10 seeds)					
FedAvg + equal-TPR thresholds	0.797	0.114	0.087	0.310	0.270
(val. split)	0.797	0.114	0.090	0.311	0.269
Naive penalty λ=1	0.716	0.076	0.102	0.231	0.233
Naive penalty λ=4	0.687	0.037	0.220	0.214	0.216
Naive penalty λ=8	0.679	0.032	0.232	0.210	0.213
Capped (20 seeds)					
FedAvg + equal-TPR thresholds	0.797	0.116	0.142	0.310	0.262
(val. split)	0.796	0.116	0.237	0.317	0.249
Naive penalty λ=1	0.719	0.080	0.125	0.232	0.239
Naive penalty λ=2	0.704	0.060	0.146	0.222	0.228
Naive penalty λ=4	0.691	0.051	0.160	0.217	0.223
Naive penalty λ=8	0.681	0.046	0.168	0.215	0.221

trained here, and the main conclusions (masking, penalty cost, thresholds, rebalancing) hold centrally as well.

Table S8: Gaps at other selection rates (DP_{int} / EO_{int}; mean over seeds; rule (q) with the top 5, 9.3, 15, 20% of scores selected); mean over 10 natural / 20 capped seeds.

Model	5%	9.3%	15%	20%
Natural				
FedAvg	0.156 / 0.307	0.254 / 0.461	0.377 / 0.619	0.468 / 0.698
Naive penalty λ=8	0.031 / 0.132	0.033 / 0.232	0.083 / 0.171	0.103 / 0.151
Oversample λ=8	0.034 / 0.119	0.047 / 0.216	0.119 / 0.123	0.173 / 0.142
Capped				
FedAvg	0.154 / 0.300	0.251 / 0.453	0.372 / 0.609	0.460 / 0.664
Naive penalty λ=8	0.031 / 0.123	0.048 / 0.172	0.085 / 0.161	0.112 / 0.131
Oversample λ=8	0.032 / 0.126	0.053 / 0.190	0.116 / 0.140	0.154 / 0.135

B.6 Robustness checks

Leave-states-out, deployable threshold (Table S15; exploratory). With whole states held out and the threshold fixed from training scores, oversampling offsets are DP -0.025 $[-0.027, -0.021]$, EO $+0.072$ $[+0.068, +0.077]$ (natural) and DP -0.002 $[-0.008, +0.004]$, EO $+0.010$ $[-0.015, +0.036]$ (capped): the capped EO effect is not distinguishable from zero here. Label balancing reproduces the natural effect (EO $+0.050$)

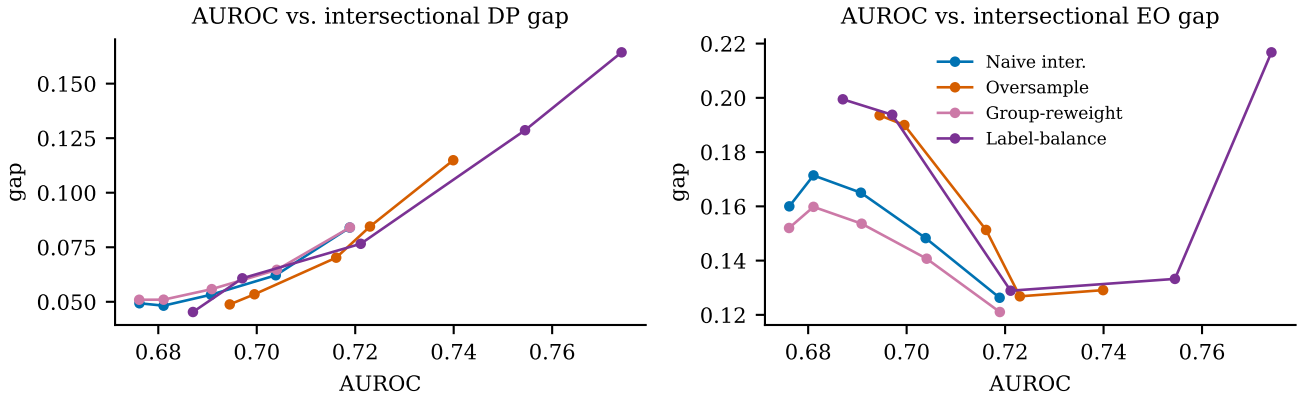


Figure 1: Accuracy–fairness frontiers (BRFSS capped, n=150; points $\lambda \in \{1, 2, 4, 8, 16\}$, mean of 20 seeds).

Table S9: Group-wise AUROC and calibration-in-the-large (mean predicted minus observed prevalence); groups: younger F, younger M, older F, older M. Mean over 20 seeds per scenario.

	AUROC				CITL			
Model	yF	yM	oF	oM	yF	yM	oF	oM
Natural								
FedAvg	0.80	0.79	0.72	0.70	−0.003	+0.009	+0.005	−0.004
Naive penalty λ=8	0.78	0.76	0.67	0.65	+0.065	+0.061	−0.004	−0.084
Oversample λ=8	0.78	0.77	0.68	0.65	+0.381	+0.386	+0.363	+0.293
Capped								
FedAvg	0.79	0.80	0.72	0.70	−0.001	+0.011	+0.007	−0.005
Naive penalty λ=8	0.77	0.76	0.67	0.65	+0.067	+0.062	−0.004	−0.083
Oversample λ=8	0.77	0.76	0.67	0.65	+0.363	+0.364	+0.312	+0.237

Table S10: Equal-λ comparison, oversample vs. naive intersectional penalty, logistic regression, 20 seeds. ΔEO is the paired difference [95% CI].

λ	Oversample		Naive penalty		ΔEO
	AUROC	EO_{int}	AUROC	EO_{int}	(os − naive)
Natural (20 seeds)					
0.5	0.723	0.157	0.751	0.100	+0.058 [+0.049, +0.066]
1	0.718	0.183	0.717	0.097	+0.086 [+0.077, +0.096]
2	0.713	0.195	0.702	0.193	+0.002 [−0.002, +0.006]
4	0.709	0.204	0.688	0.212	−0.008 [−0.012, −0.003]
8	0.706	0.210	0.680	0.222	−0.012 [−0.018, −0.007]
16	0.705	0.213	0.676	0.229	−0.016 [−0.022, −0.009]
Capped (20 seeds)					
0.5	0.759	0.161	0.751	0.129	+0.033 [+0.015, +0.052]
1	0.740	0.129	0.719	0.126	+0.003 [−0.017, +0.022]
2	0.723	0.127	0.704	0.148	−0.021 [−0.048, +0.005]
4	0.716	0.151	0.691	0.165	−0.014 [−0.043, +0.015]
8	0.700	0.190	0.681	0.171	+0.019 [−0.017, +0.052]
16	0.695	0.194	0.676	0.160	+0.034 [−0.006, +0.071]

but not the capped one (+0.017 [−0.007, +0.042]); group re-weighting lowers EO in natural (−0.006 [−0.008, −0.003]) but not significantly in capped (−0.005 [−0.013, +0.004]).

Heterogeneous clients (Tables S16, S17). These offsets use the core-grid frontier (a chord that overstates natural-data EO offsets by 0.01–0.03; Table S12), so read the comparisons between federated and centralized values, not the absolute sizes. Real states are near-IID, so we re-partitioned the same training data. Under Dirichlet α=1, oversampling behaves the same federated and centrally (EO +0.073 vs. +0.064, difference +0.009 [−0.008, +0.023]; DP −0.021 vs. −0.018), but group re-weighting with local counts does not: its EO offset is +0.016 [+0.008, +0.026] federated and −0.005 centrally (difference +0.022 [+0.012, +0.032]). Under α=0.3 oversampling's EO effect is *smaller* federated (+0.012 [−0.012, +0.036] vs. +0.066 centrally; difference −0.055 [−0.077, −0.032]) and local group re-weighting is harmful (+0.027 [+0.009, +0.047] vs. −0.005; difference +0.032 [+0.014, +0.053]). The harm comes from the local weights: when each client weights groups by its own counts, a client missing a group assigns it very different weight than the pooled data would. Re-weighting with *global* group counts (Table S17; the grid of this suite is $\lambda \in \{1, 4, 8, 16\}$, so values differ slightly from Table S16) gives an EO offset of −0.003 [−0.005, −0.001] at α=1 and −0.008 [−0.012, −0.005] at α=0.3, against +0.016 and +0.027 for local counts. So local re-weighting is a federation-specific failure under skew, and it is repaired by four pooled counts; the repaired version gives a small EO gain of the size seen on near-IID BRFSS.

Further checks on client size, age cut-off and shrinkage are in Appendix D: the oversampling EO effect appears at every client size and for age cut-offs 65 and three bands but not 45, and shrinkage gives no benefit.

C Extended discussion and limitations

Practice. (1) Audit every attribute you will report on, and their crossings where cell sizes allow; a single-attribute audit passed a model with a 12× larger intersectional gap, while an audit of both marginals would not have: the sex-only model's age-marginal DP gap (0.154) equals FedAvg's (0.152) and is six times that of the marginal-both model (0.024; Table S3). (2) Report DP, EO, sensitivity, PPV and calibration together, at more

Table S11: Frontier offsets at equal AUROC vs. the naive intersectional penalty (negative = better), logistic regression, rule (q), core grid $\lambda \in \{1, \ldots, 16\}$. Mean [95% CI over 20 seeds per scenario].

	Natural (20 seeds)		Capped, n=150 (20 seeds)	
Method	DP_{int} offset	EO_{int} offset	DP_{int} offset	EO_{int} offset
Oversample	−0.012 [−0.014, −0.010]	+0.059 [+0.054, +0.064]	−0.006 [−0.014, +0.002]	+0.020 [−0.004, +0.041]
Shrinkage	−0.000 [−0.000, −0.000]	−0.000 [−0.000, −0.000]	−0.002 [−0.002, −0.001]	+0.005 [+0.003, +0.007]

Table S12: Offsets at each λ separately (rule (q)): naive frontier AUROC and EO at that λ, then offsets at the method's own AUROC ('–' if fewer than half of the seeds fall inside the naive AUROC range: capped λ=0.5, 3 of 20 seeds). Natural: naive frontier from the dense grid $\lambda \in \{0.5, 1, 2, 4, 8, 16\}$, 20 seeds; capped: core grid. Mean [95% CI].

	Naive	Naive	Oversample offset		Group re-weight offset	
Setting, λ	AUROC	EO_{int}	DP_{int}	EO_{int}	DP_{int}	EO_{int}
Nat. 0.5	0.751	0.100	−0.011 [−0.013, −0.008]	+0.060 [+0.051, +0.069]	–	–
Nat. 1	0.717	0.097	−0.018 [−0.021, −0.016]	+0.085 [+0.076, +0.095]	+0.003 [+0.002, +0.003]	−0.023 [−0.030, −0.017]
Nat. 2	0.702	0.193	−0.015 [−0.018, −0.013]	+0.071 [+0.065, +0.077]	+0.002 [+0.002, +0.003]	−0.008 [−0.012, −0.005]
Nat. 4	0.688	0.212	−0.012 [−0.014, −0.009]	+0.057 [+0.052, +0.062]	+0.001 [+0.001, +0.002]	−0.006 [−0.008, −0.004]
Nat. 8	0.680	0.222	−0.009 [−0.011, −0.007]	+0.046 [+0.040, +0.053]	+0.001 [+0.000, +0.001]	−0.007 [−0.009, −0.005]
Nat. 16	0.676	0.229	−0.007 [−0.008, −0.005]	+0.038 [+0.031, +0.046]	–	–
Cap. 0.5	0.751	0.129	–	–	–	–
Cap. 1	0.719	0.126	−0.002 [−0.010, +0.006]	+0.001 [−0.019, +0.021]	−0.000 [−0.002, +0.002]	−0.007 [−0.014, +0.000]
Cap. 2	0.704	0.148	−0.007 [−0.015, +0.001]	−0.002 [−0.024, +0.019]	+0.002 [+0.001, +0.004]	−0.008 [−0.015, −0.001]
Cap. 4	0.691	0.165	−0.010 [−0.019, −0.001]	+0.020 [−0.007, +0.047]	+0.002 [+0.001, +0.004]	−0.011 [−0.018, −0.005]
Cap. 8	0.681	0.171	−0.007 [−0.016, +0.003]	+0.037 [+0.003, +0.070]	+0.002 [+0.001, +0.004]	−0.010 [−0.016, −0.005]
Cap. 16	0.676	0.160	−0.008 [−0.018, +0.003]	+0.037 [−0.001, +0.073]	+0.001 [−0.000, +0.003]	−0.007 [−0.014, +0.001]

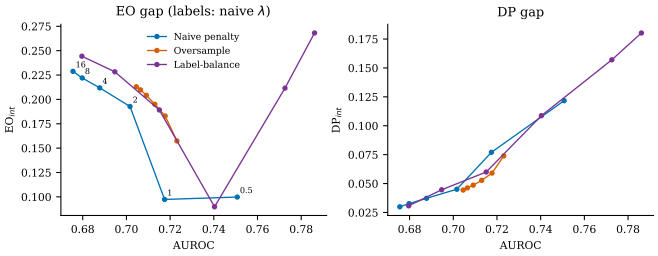


Figure 2: Natural data, dense grid $\lambda \in \{0.5, 1, 2, 4, 8, 16\}$, mean of 20 seeds. The naive EO curve is U-shaped (minimum near λ=0.5–1), so a chord between λ=1 and 4 overstates the oversampling offset.

than one operating point; a fix can improve one while worsening another, and the gaps of an unconstrained model change severalfold with the selection rate. (3) Compare mitigations at equal AUROC and across penalty strengths: the EO gap is U-shaped in λ, and a group-threshold baseline that looks clearly better against a strong penalty only ties a weak one. (4) Do not credit a federated method for an effect a centralized control reproduces, and test it under client heterogeneity; local-count re-weighting is the one mitigation here that federation breaks. (5) Train to convergence, with a schedule matched to penalty stiffness; our own earlier under-trained runs gave different answers.

Privacy cost of thresholds. Group thresholds need the sensitive attribute at decision time, and fitting them needs pooled group-wise scores: each client would have to share, per group, score quantiles or histograms with a server. With hospital-scale clients the cells are tiny (with 150 respondents per client, the younger-women cell holds on the order of 40 respondents and about one positive on average), so the shared statistics can approach patient-level information, and a secure-aggregation or differentially private histogram would add noise to exactly the small groups the thresholds are meant to protect. We fitted thresholds centrally on training scores or on a pooled validation split, and did not measure the privacy–accuracy cost of a federated estimator; we therefore make no claim about what a private federated procedure would achieve.

Limitations. (1) *Data.* One self-reported US survey with near-IID states [3]; federated claims under skew rest on synthetic Dirichlet partitions; two binary attributes only, so the value of crossing three or more attributes (for example race or income, which the cleaned extract used here does not contain) is untested and is future work; the marginal-sufficiency result may not carry over. (2) *Models.* Logistic regression and a 16-unit MLP whose penalized training is optimization-sensitive (Sec. H); MLP results are exploratory. FairFed, FedIDA and exponentiated gradient are not benchmarked here. (3) *Convergence.* The schedule is chosen by penalty stiffness and checked against half the rate and twice the rounds on a subset of seeds and arms (Appendix E), not for every run. (4) *Statistics.* Seeds are random splits of one dataset (5–20 per suite, listed in each caption); the

Table S13: Centralized matched control vs. federated: frontier offsets, logistic regression, rule (q), mean [95% CI over seeds]. "Diff" = federated − central (paired); 20 seeds per scenario; natural: dense frontier $\lambda \in \{0.5, 1, 2, 4, 8, 16\}$, capped: core grid.

	Natural			Capped		
Variant	Central	Federated	Diff	Central	Federated	Diff
Oversample (DP)	−0.016 [−0.018, −0.014]	−0.012 [−0.014, −0.010]	+0.004 [+0.004, +0.005]	−0.008 [−0.016, +0.000]	−0.008 [−0.017, +0.001]	−0.001 [−0.006, +0.004]
Oversample (EO)	+0.052 [+0.047, +0.057]	+0.059 [+0.054, +0.064]	+0.008 [+0.005, +0.010]	+0.025 [+0.001, +0.050]	+0.032 [+0.000, +0.062]	+0.009 [−0.014, +0.031]
Label-balance (dup.) (DP)	−0.005 [−0.006, −0.004]	−0.002 [−0.003, −0.001]	+0.003 [+0.003, +0.003]	−0.007 [−0.012, −0.003]	−0.006 [−0.014, +0.002]	+0.002 [−0.004, +0.007]
Label-balance (dup.) (EO)	+0.026 [+0.023, +0.029]	+0.029 [+0.026, +0.032]	+0.003 [+0.002, +0.005]	+0.026 [+0.012, +0.039]	+0.030 [−0.003, +0.061]	+0.004 [−0.018, +0.027]
Group re-weight (DP)	+0.001 [+0.001, +0.001]	+0.002 [+0.001, +0.002]	+0.001 [+0.000, +0.001]	+0.001 [+0.000, +0.002]	+0.002 [+0.001, +0.003]	+0.001 [−0.000, +0.002]
Group re-weight (EO)	−0.007 [−0.009, −0.005]	−0.011 [−0.013, −0.009]	−0.004 [−0.005, −0.003]	−0.003 [−0.006, −0.000]	−0.010 [−0.015, −0.004]	−0.007 [−0.012, −0.002]

Table S14: Control experiments for the oversampling EO effect (exploratory): offsets vs. the dense naive frontier at equal AUROC, mean [95% CI over 20 seeds per scenario]. Upper rows use the core λ points for every arm; the last three rows use the dense natural grid ($\lambda \in \{0.5, \ldots, 16\}$).

	Natural (20 seeds)		Capped (20 seeds)	
Variant	DP_{int}	EO_{int}	DP_{int}	EO_{int}
Oversample (cells, noise)	−0.011 [−0.013, −0.009]	+0.056 [+0.051, +0.061]	−0.007 [−0.015, +0.001]	+0.024 [−0.000, +0.047]
orig. n_k in FedAvg	−0.011 [−0.013, −0.009]	+0.056 [+0.051, +0.061]	−0.007 [−0.015, +0.001]	+0.025 [+0.000, +0.048]
no noise (pure duplication)	−0.011 [−0.013, −0.009]	+0.056 [+0.052, +0.061]	−0.008 [−0.015, +0.000]	+0.027 [+0.005, +0.047]
Label-balance by duplication only	−0.004 [−0.005, −0.002]	+0.042 [+0.038, +0.046]	−0.008 [−0.015, −0.002]	+0.020 [−0.008, +0.045]
Loss reweight: labels only	−0.006 [−0.007, −0.006]	+0.030 [+0.026, +0.034]	−0.007 [−0.009, −0.004]	+0.012 [+0.005, +0.021]
Loss reweight: (group,label) cells	−0.003 [−0.003, −0.002]	+0.027 [+0.024, +0.031]	−0.006 [−0.012, −0.000]	+0.020 [+0.004, +0.034]
Loss reweight: groups only	+0.001 [+0.001, +0.002]	−0.012 [−0.014, −0.009]	+0.002 [+0.000, +0.003]	−0.010 [−0.016, −0.005]
Oversample, dense natural grid	−0.012 [−0.014, −0.010]	+0.059 [+0.054, +0.064]	–	–
Label-balance, dense natural grid	−0.002 [−0.003, −0.001]	+0.029 [+0.026, +0.032]	–	–
Loss reweight: groups only, dense natural grid	+0.002 [+0.001, +0.002]	−0.011 [−0.013, −0.009]	–	–

Holm-adjusted floor is about 0.02 for 10 seeds, and with 5 seeds (MLP natural) no test can be significant, so small effects there are inconclusive rather than absent; exploratory tests are uncorrected. (5) *Metrics.* DP is a parity audit, not a clinical objective; gaps depend on the operating point (Sec. B.3) and, for offsets, on the λ grid; λ is swept, not tuned. (6) *Thresholds.* They use the sensitive attribute at decision time, and the cost of estimating them privately is unmeasured.

D Additional robustness checks

Client size and age cut-off. The oversampling EO effect appears at every client size (EO $+0.029$ $[+0.004, +0.052]$ at n=75, $+0.032$ $[0.000, +0.062]$ at 150, $+0.040$ $[+0.014, +0.063]$ at 300, $+0.071$ natural), so it is not an artifact of tiny cells. It is absent for an age cut-off of 45 (EO $+0.003$ $[-0.005, +0.011]$ natural, $+0.000$ $[-0.007, +0.007]$ capped) and present for 65 ($+0.096$, $+0.056$) and for three age bands ($+0.090$; capped $+0.027$ $[-0.001, +0.055]$) (Table S18). The offset rises with the cut-off in both settings, which fits the larger prevalence spread at higher cut-offs, but this is three points and we claim no mechanism.

Shrinkage gives no benefit. Offsets are 0.000 in natural data (every client–group cell is large, so shrinkage is numerically the naive penalty) and, in capped data, DP -0.002 $[-0.003, -0.001]$ with EO $+0.005$ $[+0.003, +0.007]$ (leave-states-out: $+0.006$ $[+0.003, +0.008]$): a statistically detectable but practically negligible *worsening* of EO. We report it as no benefit, not as a null.

E Convergence of penalized models

For each penalty type and $\lambda \in \{1, 8, 16\}$ we compared the production schedule of Sec. III-A with half the learning rate and twice the rounds, without the divergence guard (Table S19; absolute differences, mean over settings; capped seeds 42–46 for all arms; natural seeds 42–46 for intersectional, oversampling, label-balance and group re-weighting at $\lambda \in \{8, 16\}$ and for the EO penalty at $\lambda \in \{1, 8\}$). In natural data, the setting that carries the headline oversampling result, the schedule does not matter: the largest EO difference over 40 DP-type and 10 EO-penalty comparisons is 0.005 and 0.007, an order of magnitude below the effect sizes (0.03–0.07). The sex-only, age-only and marginal-both arms were checked in capped data only. For the DP-type penalties the mean differences are small (capped: AUROC 0.002, DP 0.003, EO 0.009; oversampling alone 0.006, 0.006 and 0.020, the other DP-type arms 0.001, 0.002 and 0.007) but not negligible per seed: the largest EO difference is 0.111, for oversampling at $\lambda \geq 8$, where the production schedule gave the higher EO. For the EO penalty the schedule matters: unguarded, two fits (seed 45, λ=8 and 16) collapsed to AUROC 0.50, which alone puts the mean differences at 0.03 AUROC and 0.13 in DP and EO (maximum 0.74); without them the means are 0.004 AUROC, 0.019 DP and 0.031 EO, so the EO penalty is more schedule-sensitive than the DP-type penalties (0.003 DP, 0.009 EO). Converged-schedule results for the EO penalty at $\lambda \geq 8$ are therefore indicative only.

Table S15: Leave-states-out, deployable threshold: offsets vs. the naive penalty on the core-grid chord (logistic regression, 10 natural / 20 capped partitions), mean [95% CI]. Exploratory.

	Natural		Capped	
Variant	DP_{int}	EO_{int}	DP_{int}	EO_{int}
Oversample	−0.025 [−0.027, −0.021]	+0.072 [+0.068, +0.077]	−0.002 [−0.008, +0.004]	+0.010 [−0.015, +0.036]
Label-balance (dup.)	−0.012 [−0.015, −0.009]	+0.050 [+0.046, +0.054]	−0.006 [−0.011, −0.001]	+0.017 [−0.007, +0.042]
Loss re-weight: labels	−0.012 [−0.013, −0.011]	+0.038 [+0.033, +0.044]	−0.008 [−0.010, −0.005]	+0.021 [+0.010, +0.034]
Loss re-weight: cells	−0.010 [−0.012, −0.009]	+0.037 [+0.030, +0.044]	−0.009 [−0.013, −0.006]	+0.013 [−0.004, +0.031]
Loss re-weight: groups	+0.002 [+0.001, +0.002]	−0.006 [−0.008, −0.003]	+0.002 [−0.000, +0.004]	−0.005 [−0.013, +0.004]
Shrinkage	−0.000 [−0.000, −0.000]	+0.000 [−0.000, +0.000]	−0.002 [−0.002, −0.001]	+0.006 [+0.003, +0.008]

Table S16: Dirichlet-skewed client partitions of the same training data (10 seeds): offsets vs. the naive penalty on the core-grid chord (which overstates natural EO offsets, Sec. B.4), rule (q), mean [95% CI]. Federated, centralized, and their paired difference. Exploratory.

		DP_{int}			EO_{int}		
Partition	Variant	Fed.	Central	Diff	Fed.	Central	Diff
Dirichlet α=1	Oversample	−0.021 [−0.025, −0.017]	−0.018 [−0.021, −0.016]	−0.003 [−0.006, +0.000]	+0.073 [+0.061, +0.085]	+0.064 [+0.057, +0.072]	+0.009 [−0.008, +0.023]
Dirichlet α=1	Label-balance	−0.018 [−0.022, −0.013]	−0.008 [−0.010, −0.006]	−0.010 [−0.014, −0.006]	+0.067 [+0.058, +0.076]	+0.046 [+0.041, +0.050]	+0.021 [+0.013, +0.031]
Dirichlet α=1	Group re-weight	−0.005 [−0.007, −0.003]	+0.001 [+0.001, +0.002]	−0.006 [−0.008, −0.004]	+0.016 [+0.008, +0.026]	−0.005 [−0.008, −0.003]	+0.022 [+0.012, +0.032]
Dirichlet α=0.3	Oversample	−0.016 [−0.021, −0.011]	−0.018 [−0.021, −0.016]	+0.002 [−0.002, +0.007]	+0.012 [−0.012, +0.036]	+0.066 [+0.059, +0.075]	−0.055 [−0.077, −0.032]
Dirichlet α=0.3	Label-balance	−0.016 [−0.020, −0.012]	−0.008 [−0.010, −0.007]	−0.008 [−0.012, −0.003]	+0.042 [+0.027, +0.056]	+0.046 [+0.042, +0.051]	−0.004 [−0.018, +0.009]
Dirichlet α=0.3	Group re-weight	−0.007 [−0.014, −0.000]	+0.001 [+0.001, +0.002]	−0.008 [−0.015, −0.001]	+0.027 [+0.009, +0.047]	−0.005 [−0.008, −0.003]	+0.032 [+0.014, +0.053]

Table S17: Group re-weighting with local vs. global group counts under Dirichlet partitions (federated, 10 seeds): offsets vs. naive penalty, mean [95% CI]. Exploratory.

Partition	Weights	DP_{int}	EO_{int}
Dirichlet α=1	local group counts	−0.005 [−0.007, −0.003]	+0.016 [+0.008, +0.026]
Dirichlet α=1	global group counts	+0.001 [+0.001, +0.002]	−0.003 [−0.005, −0.001]
Dirichlet α=0.3	local group counts	−0.007 [−0.014, −0.000]	+0.027 [+0.009, +0.047]
Dirichlet α=0.3	global group counts	+0.002 [+0.001, +0.002]	−0.008 [−0.012, −0.005]

F Exponentiated-gradient reference

We ran an exponentiated-gradient reduction [6] with class-balanced error (plain 0–1 error collapses to an all-negative classifier at 9.3% prevalence), DP or EO constraints on the four groups and $\epsilon \in \{0.05, 0.02, 0.01\}$, centralized on pooled training data (30K-row subsample in the natural setting; class balancing by resampling for Fairlearn, by weights for our own implementation). We report two implementations, our own and Fairlearn 0.15 (run by the authors externally). Both output a randomized classifier, and we report its expected rates. The per-ϵ results (24 rows; released with the code) are illustrative only: the objective is class-balanced, so the classifier selects about 38–44% of respondents (prevalence 9.3%) and operates far from the 9.3% rule used elsewhere, and its AUROC is that of the mixture score. The two implementations disagree (natural, DP constraint at ϵ=0.01: EO_{int} 0.118 versus 0.210). We did not investigate the cause; the oracle, the number of iterations, the Lagrange-multiplier bound and class balancing by weights versus resampling all differ, and a randomized mixture is sensitive to them. EG numbers should therefore not be used as a benchmark here. We draw no comparison with the other methods.

G UCI Heart Disease (secondary, 851 patients)

On UCI (four hospitals; stratified 5-fold CV repeated 20 times; rule (q); bootstrap CIs over patients; the same stiffness-aware training schedule and guard as for BRFSS) the sex-only model at λ=8 reduces sex DP and EO by 63% and 75% but age×sex gaps by 32% and 33% (FedAvg AUROC 0.878, sex-only 0.852). Oversampling offsets are not reported: its AUROC at λ=8 (0.736) is far below the naive penalty's (0.829), so offsets would rest on two frontier points. With subgroup cells of 2–6 patients per client, UCI is too small to discriminate mechanisms; its intervals omit training variability.

H A trained MLP (exploratory)

The MLP (16 tanh units) uses T=400 rounds at rate 0.1 and reaches the same unpenalized AUROC as logistic regression (0.797 natural, 0.795 capped). Penalized MLPs are the least reliable models in this study. In a single-seed check (capped, seed 42, λ=8) the intersectional DP gap was 0.091, 0.125 and 0.135 after 50, 150 and 400 rounds at rate 0.1–0.25, and EO_{int} 0.074, 0.116 and 0.133, so the earlier T=50 MLP overstated what the penalty achieves; at T=400 a rate of 0.25 instead of 0.1 still gave EO_{int} 0.208 against 0.133 at nearly identical AUROC (0.686 vs. 0.687). We therefore treat all MLP results as exploratory and do not draw conclusions from small differences.

With that caveat (Tables S20, S21): the sex-only MLP shows the same masking pattern as logistic regression (natural: sex DP 0.125 → 0.045, −64%, but DP_{int} 0.266 → 0.196, −26%; capped DP_{int} 0.260 → 0.190, −27%). The intersec-

Table S18: Age cut-offs (exploratory). Offsets vs. the naive penalty of the same configuration at equal AUROC, mean [95% CI].

	Natural (10 seeds)		Capped (20 seeds)	
Variant	DP_{int}	EO_{int}	DP_{int}	EO_{int}
Age cut 45: oversample	−0.005 [−0.007, −0.003]	+0.003 [−0.005, +0.011]	−0.002 [−0.006, +0.001]	+0.000 [−0.007, +0.007]
Age cut 45: shrinkage	−0.000 [−0.000, +0.000]	−0.000 [−0.000, −0.000]	−0.001 [−0.002, −0.001]	−0.001 [−0.004, +0.001]
Age cut 65: oversample	−0.037 [−0.041, −0.034]	+0.096 [+0.090, +0.103]	−0.014 [−0.024, −0.002]	+0.056 [+0.026, +0.086]
Age cut 65: shrinkage	−0.000 [−0.000, +0.000]	+0.000 [−0.000, +0.000]	−0.001 [−0.002, +0.001]	+0.005 [+0.002, +0.008]
3 age bands × sex: oversample	+0.012 [+0.009, +0.014]	+0.090 [+0.074, +0.107]	−0.012 [−0.018, −0.005]	+0.027 [−0.001, +0.055]

Table S19: Production schedule vs. half learning rate / double rounds (unguarded): mean absolute difference in AUROC, DP_{int}, EO_{int}; n = settings; last column: largest EO difference.

Scenario	Penalty	n	$\overline{\lvert\Delta\rvert}$ AUROC	$\overline{\lvert\Delta\rvert}$ DP_{int}	$\overline{\lvert\Delta\rvert}$ EO_{int}	max $\lvert\Delta\rvert$ EO_{int}
capped	DP	90	0.0016	0.0028	0.0089	0.111
capped	EO (excl. collapsed fit)	13	0.0035	0.0188	0.0308	0.070
capped	EO	15	0.0310	0.1284	0.1256	0.743
natural	DP	40	0.0002	0.0004	0.0009	0.005
natural	EO	10	0.0001	0.0006	0.0019	0.007
capped	DP: oversample	15	0.0061	0.0056	0.0196	0.111
capped	DP: other DP arms	75	0.0007	0.0022	0.0067	0.039
natural	DP: oversample	10	0.0004	0.0003	0.0011	0.005
natural	DP: other DP arms	30	0.0001	0.0004	0.0009	0.005

tional penalty is *less* effective on DP for the MLP than for logistic regression (DP_{int} 0.087 natural, 0.096 capped at λ=8, against 0.033 and 0.048), but its EO gap is lower (0.164 and 0.157). Oversampling lowers DP_{int} more than for logistic regression (-0.078 $[-0.102, -0.045]$ natural, 5 seeds; -0.039 $[-0.061, -0.018]$ capped, 10 seeds). Its EO effect is positive in natural data ($+0.048$ $[-0.005, +0.096]$; label balancing $+0.056$ $[+0.021, +0.098]$) and null in capped data ($+0.003$ $[-0.054, +0.047]$). The MLP tests are exploratory and not part of the Holm family: natural has 5 seeds, for which the smallest attainable p is 0.0625, so no natural MLP test can be significant at 0.05, and capped intervals are unadjusted. Group re-weighting lowers EO in natural data (-0.013 $[-0.020, -0.008]$) and not detectably in capped data. The MLP therefore neither confirms nor contradicts the logistic-regression EO effect of oversampling.

Reproducibility

Code, cleaned BRFSS/UCI tables and per-seed results for every table are provided. Experiments: `run_brfss.py`, `run2.py`, `engine.py`, `fairlearn_eg.py`, `run_uci_v3.py` (UCI), `agegap.py`, `agecoef.py`, `convcheck2.py`, `convcheck3.py`, `convcheck4.py`; batch scripts `run6.sh`–`run16.sh` (run with `CONV=2` where noted). Tables and figures: `analysis.py`, `analysis2.py`, `make_results.py`, `make_tables.py`, `make_tables2.py`, `nums3.py`, `nums4.py`, `nums5.py`, `dense.py`, `fig_dense.py`, `checks3.py` (verifies numbers quoted in the text). Seeds per suite are listed in Table S1. The supplementary material (Sections A–F, Tables S1–S10: additional tables, robustness checks, MLP, convergence, exponentiated-gradient reference, UCI replication) is submitted with the paper and also in the repository (`supplement.pdf`). Code: `github.com/pradhuman001/fedcvd-fairness`.

Table S20: MLP (16 tanh units, T=400, rate 0.1) at λ=8; mean over seeds (5 natural, 10 capped). Exploratory.

Method	AUROC	DP_{sex}	DP_{int}	EO_{sex}	EO_{int}
Natural (5 seeds)					
FedAvg	0.797	0.125	0.266	0.312	0.477
Sex-only	0.791	0.045	0.196	0.096	0.365
Naive inter.	0.693	0.020	0.087	0.036	0.164
Oversample	0.712	0.012	0.042	0.029	0.228
Group re-weight	0.691	0.022	0.083	0.043	0.141
Capped (10 seeds)					
FedAvg	0.795	0.117	0.260	0.285	0.463
Sex-only	0.787	0.039	0.190	0.083	0.354
Naive inter.	0.688	0.027	0.096	0.057	0.157
Oversample	0.703	0.023	0.079	0.063	0.161
Group re-weight	0.687	0.027	0.093	0.056	0.156

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Table S21: MLP frontier offsets vs. naive penalty (rule (q)), mean [95% CI over seeds]; 5 natural, 10 capped seeds. Exploratory.

Method	Natural DP_{int}	Natural EO_{int}	Capped DP_{int}	Capped EO_{int}
Oversample	-0.078 [-0.102, -0.045]	$+0.048$ [-0.005, $+0.096$]	-0.039 [-0.061, -0.018]	$+0.003$ [-0.054, $+0.047$]
Label-balance (dup.)	-0.049 [-0.077, -0.019]	$+0.056$ [$+0.021$, $+0.098$]	-0.042 [-0.062, -0.022]	$+0.019$ [-0.034, $+0.066$]
Group re-weight	-0.002 [-0.004, $+0.001$]	-0.013 [-0.020, -0.008]	-0.003 [-0.006, $+0.000$]	-0.005 [-0.013, $+0.002$]

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